Towards a High-Resolution Local Tomography Using Statistical Iterative Reconstruction

Essam A. Rashed, *Member, IEEE,* Hiroyuki Toda, Toshihiro Sera, Akira Tsuchiyama, Tsukasa Nakano, Kentaro Uesugi and Hiroyuki Kudo, *Member, IEEE*

***Abstract*—Synchrotron radiation (SR) x-ray micro-CT is an effective method for high-resolution imaging of small objects with several applications in biology and industry. However, the detector field of view is tiny, which limits the sample size to a few millimeters. When the sample size is larger than the limited field of view, reconstructed images, using conventional methods, known to suffer from DC-shift and low-frequency artifacts. This problem is known as local tomography or interior problem. In this paper we introduce a statistical iterative image reconstruction method to eliminate image artifacts produced from local tomography. The proposed method can be used in several SR imaging applications to enable a high resolution of the scanned object while preserving the image quality from artifacts produced due to the local tomography problem.**

I. Introduction

Synchrotron radiation (SR) micro-CT systems that provide a high-resolution cross-sectional images of small animals and other applications had a strong impact in several research fields such as drug developments. To obtain high-resolution CT images, it is required that the object is located completely inside the detector field of view (FOV). However, in many cases, a portion of the scanned object lies outside the FOV which leads to a truncation in the acquired projection data. Data truncation may also occur due to the magnification of a specific portion of the object or to reduce unnecessary radiation to the remaining portion of the object (Fig. 1). The most sever truncation case is when the projection data is truncated in all view angles providing the so-called local tomography or the interior problem [1]. It is known that when the projection data is fully truncated such as the case of local CT imaging, the reconstructed images suffers from DC-shift (or cupping) and low-frequency artifacts. These artifacts become strong in regions close to the boundaries of the FOV. The local tomography is also known as an ill-posed inverse problem which had been studied for a long time. Based on the analytical reconstruction, it was believed that all projection rays passing through the whole object are required for exact reconstruction and any truncation leads to losing the solution exactness. However, the recently developed theories based on the concept of Differentiated Backprojection (DBP) [2]–[4] and Backprojection-Filtration (BPF) [5] succeeded in reducing the required set of projection rays for theoretically exact and stable reconstruction.

This work was partially supported by Grant-in-Aid from the Japan Society for the Promotion of Science (JSPS). Grant for foreign post-doctoral fellows (ID No. P10052).

E.A.Rashed and H.Kudo are with the Department of Computer Science, Graduate School of Systems and Information Engineering, University of Tsukuba, Tsukuba 305-8573, Japan. H.Toda is with the Toyohashi University of Technology, Toyohashi, Japan, T.Sera and A.Tsuchiyama are with the Osaka University, Osaka, Japan. T.Nakano is with the AIST, Tsukuba, Japan. K.Uesugi is with the Japan Synchrotron Radiation Research Institute, Hyogo, Japan. E.A.Rashed is also with the Department of Mathematics, Faculty of Science, Suez Canal University, Ismailia 41522, Egypt (e-mail: essam@imagelab.cs.tsukuba.ac.jp).

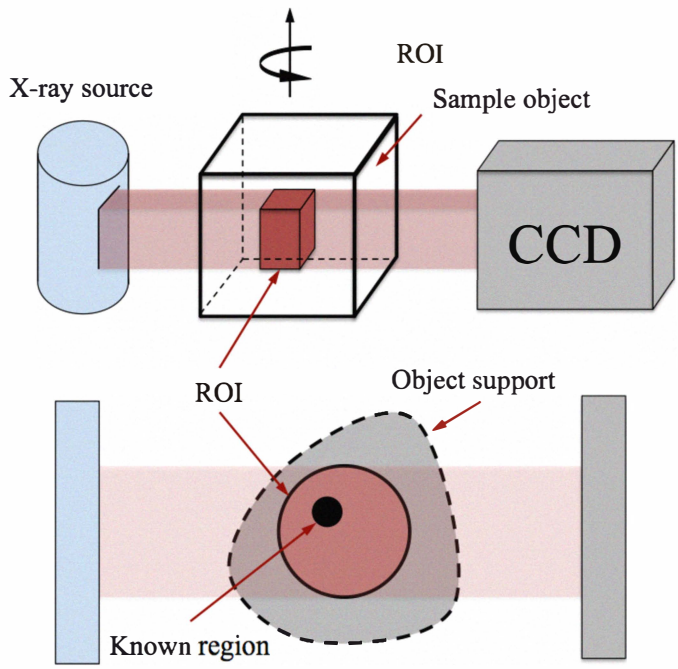

Fig. 1. Local tomography in SR micro-CT. The sample object usually extend outside the FOV of the CCD used for data acquisition. For accurate reconstruction, it required to know the object contour line (object support in additional to a region inside the region of interest (ROI).

In our earlier work, we proposed a statistical iterative reconstruction algorithm for region-of-interest (ROI) reconstruction when the projection data is partially truncated [6]. Here, we extend this algorithm to the case of local tomography. Experimental results using real data obtained from high-resolution SR imaging indicates a significant reduction in the image artifacts when the proposed method is used.

II. Proposed Method

A. Object support constraint

According to the theories based on the analytical concept of Hilbert transform in [3], [4], [7], it is possible to reconstruct exact image from locally truncated data if the following constrains are satisfied. 1) All projection rays passing through

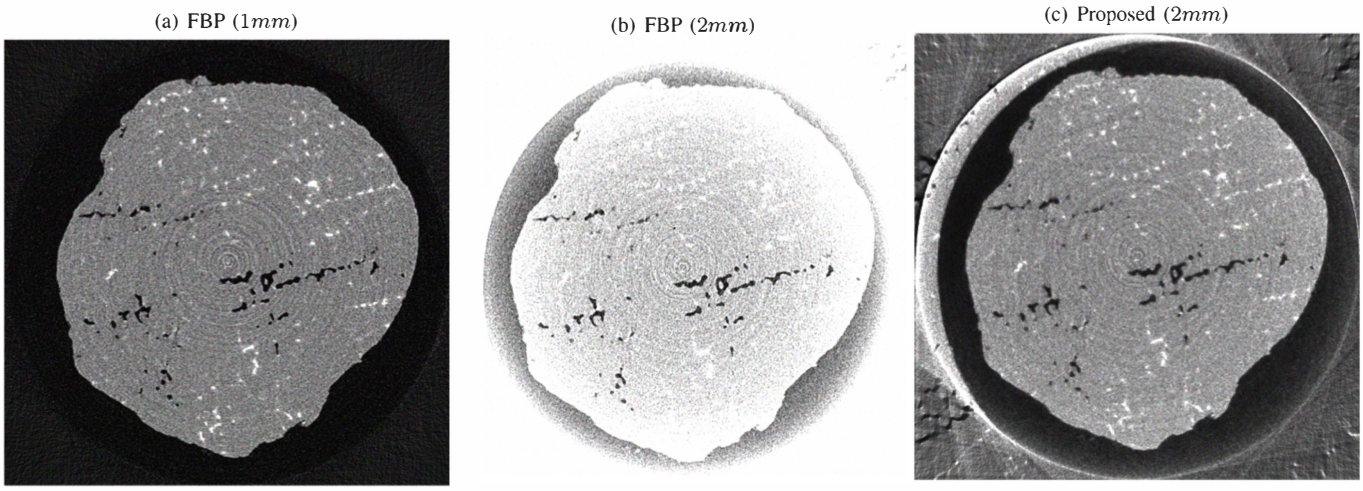

Fig. 2. Image reconstruction for the Aluminum alloy data. (a) Image reconstructed using FBP method from non-truncated data, to be used as true reference. Images reconstructed from local data using FBP algorithm in (b) and proposed method in (c).

FOV are included in the measured projection data, 2) a compact support of the object is available, 3) *a priori* known information of intensity value for small region inside the FOV is available, and 4) for iterative reconstruction, the image matrix should be large enough to include the entire object. The second constraint is not always available, as the exact definition of the object contour requires additional efforts such as extra hardware. The iterative reconstruction algorithm for ROI reconstruction (ROI-OSEM), detailed in [6], provides an automatic and efficient detection of the object support in parallel to the image reconstruction. In the following, we review the main approach of the ROI-OSEM algorithm with modifications that suits the correct noise modeling in transmission CT:

B. Iterative reconstruction algorithm

We consider the following noise model:

$$y_i \approx \text{Poisson}\ [b_i \exp(-\langle \vec{a_i}, \vec{x} \rangle)] \tag{1}$$

where $\vec{x} = (x_1, \ldots, x_n)^\top$ is the image vector, $\vec{y} = (y_1, \ldots, y_m)^\top$ be the measured counts, $\vec{b} = (b_1, \ldots, b_m)$ is the blank scan, $A = \{a_{ij}\}$ is the $m \times n$ system matrix and $\langle ., . \rangle$ is the regular inner product. The cost function for image reconstruction is defined as:

$$f(\vec{x}) = L(\vec{x}) + \beta T(\vec{x}), \tag{2}$$

where $L(\vec{x})$ is the negative log-likelihood function, $T(\vec{x})$ is a penalty function based on ℓ_0 norm and β is a hyperparameter defined as:

$$L(\vec{x}) = \sum_{i=1}^{m} \left[b_i \exp(-\langle \vec{a_i}, \vec{x} \rangle) + y_i \langle \vec{a_i}, \vec{x} \rangle \right], \tag{3}$$

$$T(\vec{x}) = \sum_{j=1}^{n} s(x_j), \quad s(t) = \lim_{\epsilon \to 0} |t|^\epsilon \equiv \begin{cases} 1 & t \neq 0 \\ 0 & t = 0 \end{cases}. \tag{4}$$

The formulation of the image reconstruction cost function in Eq. (2) can be simply indicate that, the reconstructed image is the most sparse image that fit with the measured projection data. This approach can successfully find the object support by threshold the image pixels that is close to zero to the value of zero. The minimization of cost function using the majorization-maximization approach [8] leads to the following iterative reconstruction algorithm:

1 Initialization $x^{(0)} = \epsilon$, with $\epsilon > 0$, $k \leftarrow 0$
repeat
  for $j = 1 : n$ **do**
2 $p_j = x_j^k + x_j^k \frac{\sum_i a_{ij}\left[b_i \exp(-\langle \vec{a_i}, \vec{x}^k \rangle) - y_i\right]}{\sum_i a_{ij} \langle \vec{a_i}, \vec{x}^k \rangle b_i \exp(-\langle \vec{a_i}, \vec{x}^k \rangle)}$
    $t_j = \frac{\sum_i a_{ij} \langle \vec{a_i}, \vec{x}^k \rangle b_i \exp(-\langle \vec{a_i}, \vec{x}^k \rangle)}{2\beta x_j^k}$
3 $x_j^{k+1} = \begin{cases} 0 & (p_j \leq \sqrt{1/t_j}) \\ p_j & (otherwise) \end{cases}$
  end
  $k \leftarrow k + 1$
until *stopping criterion*;

III. Experimental Results

In the experiment studies, we have used a three-dimensional high-resolution CT data obtained from a third-generation synchrotron radiation source located in Hyogo, Japan (SPring-8) [9].

A. Aluminum alloy data

In the first experiment, the object sample was a metallic Aluminum alloy (Al-Cu). In this imaging application, the air holes and the distribution of the alloy components are required to be observed in the sample. Two sets of data are used, the first one was acquired using sample of size $1mm$, which is completely fit inside the FOV to be used as a true reference for the later scan. The second dataset was acquired using a sample of size $2mm$ by including an external shield to the original sample imaged earlier. The data is acquired from BL20XU beamline with CCD of 2000×1312 that reconstructed in

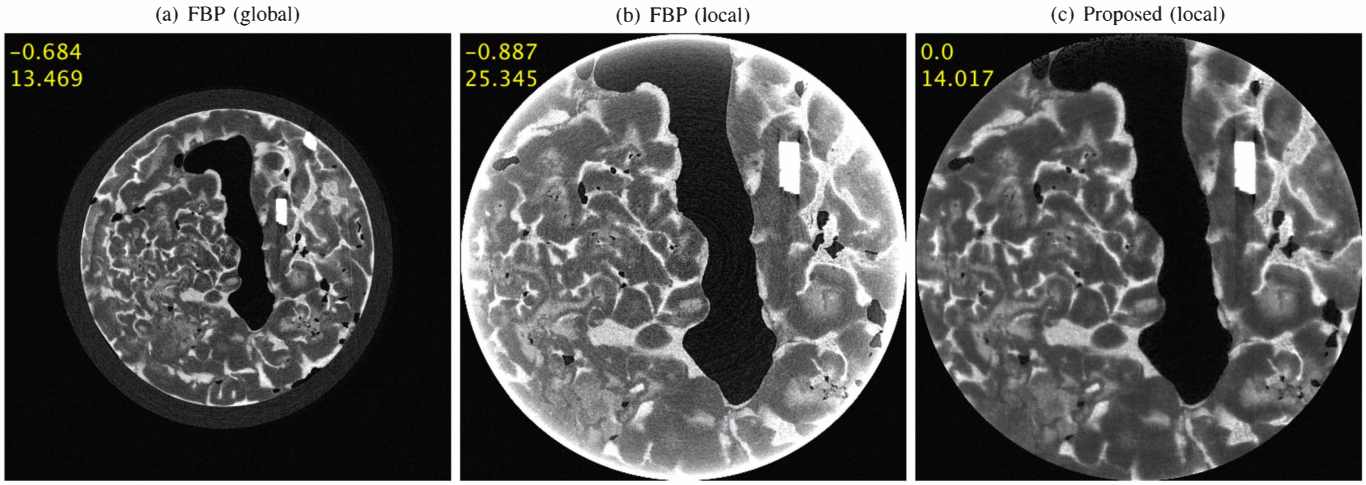

Fig. 3. Global image reconstructed from full projection data (left), local images reconstructed using standard FBP algorithm (middle), and ROI-OSEM algorithm (right). Intensity value ranges are displayed on top left corner of each images.

volume of size $2000 \times 2000 \times 1312$ with pixel size of $0.5\mu m$. The projection data was measured for 1500 views over 180° with energy of 35 keV. A single slice of the first dataset, which is not truncated, was reconstructed using FBP algorithm. The corresponding slice in the second dataset was reconstructed using FBP and proposed method. Results, shown in Fig. 2, indicate that the proposed method can reduce the DC-shift and low-frequency artifacts in the image reconstructed from local data.

B. Calcite in tarc

In this experiment, the object was a sample of tarc fluid containing calcite powder. In this application, it is required to investigate the internal structure of micro-size sample. The exact intensity value is important to identify the correct structure and any DC-shift occurs due to data truncation may lead to incorrect identification of the sample contents. Two sets of data, global and local, were acquired using an x-ray energy of 20 keV at BL20B2 beamline. The global data was obtained using a detector size of 1560×680 bins with bin size of 6.66 μm. In these settings the whole object was located completely inside the detector FOV and the projection data was computed over 2250 view angles over 180°. The local data was acquired with a magnification of the central region of the sample. The data was obtained using a detector of size 2000×1312 bins with bin size of 3.33 μm and 3000 view angels over 180°. Although, the local data provides a higher resolution (doubled) to the central region of the sample, the projection data was fully truncated and represents a local tomography.

The global data corresponding to central slice was used to reconstruct image of the whole object with 1560×1560 pixels using standard FBP algorithm, shown in Fig. 3(a) to be used as a reference for the evaluation of the proposed method. The local data of the central slice was used to reconstruct a magnified image of 2000×2000 pixels of an internal structure of the sample. The reconstruction using standard FBP and proposed method with 10 iterations are shown in Fig. 3. The proposed method succeeded in reducing the artifacts existing in the analytical reconstruction from local projection data. The correct estimation of the object support and the internal region, corresponding to air hole, lead to an exact local reconstruction.

C. Mouse lung

In this experiment, a real data acquired from a mouse, was used to investigate the structure of lung alveoli. The mouse was imaged BL20B2 beamline. The projection data was acquired using a detector of 2000×300 bins with bin size of 3.33 μm with a set of 1500 projection angles over 180°. The projection data was fully truncated such that the detector FOV was located completely inside the mouse. No information regarding the true object were measured during data acquisition. A single slice was reconstructed using standard FBP and the proposed method. The reconstructed images and the corresponding central horizontal and vertical profiles are shown in Fig. 4.

IV. Conclusion

In this work, we proposed statistical iterative reconstruction algorithm for local tomography. The proposed method is based on simple approach that effectively estimate the require *a priori* known information for accurate and stable reconstruction from fully truncated projection data. The proposed method is tested using SR micro-CT real data and effective achievement in image quality is observed compared to conventional methods. However, the computation cost of the proposed method still the major challenge. Future work include the reducing the computation cost to a reasonable level for practical implementation in high-resolution micro-CT imaging.

References

[1] F. Natterer, *The Mathematics of Computerized Tomography*, Philadelphia, PA: Wiley, 1986.

[2] F. Noo, R. Clackdoyle and J. D. Pack, "A two-step Hilbert transform method for 2D image reconstruction," *Phys. Med. Biol.*, vol. 49, pp. 3903-3923, 2004.

[3] M. Defrise, F. Noo, R. Clackdoyle and H. Kudo, "Truncated Hilbert transform and image reconstruction from limited tomographic data," *Inverse Problems*, vol. 22, pp. 1037-1053, 2006.

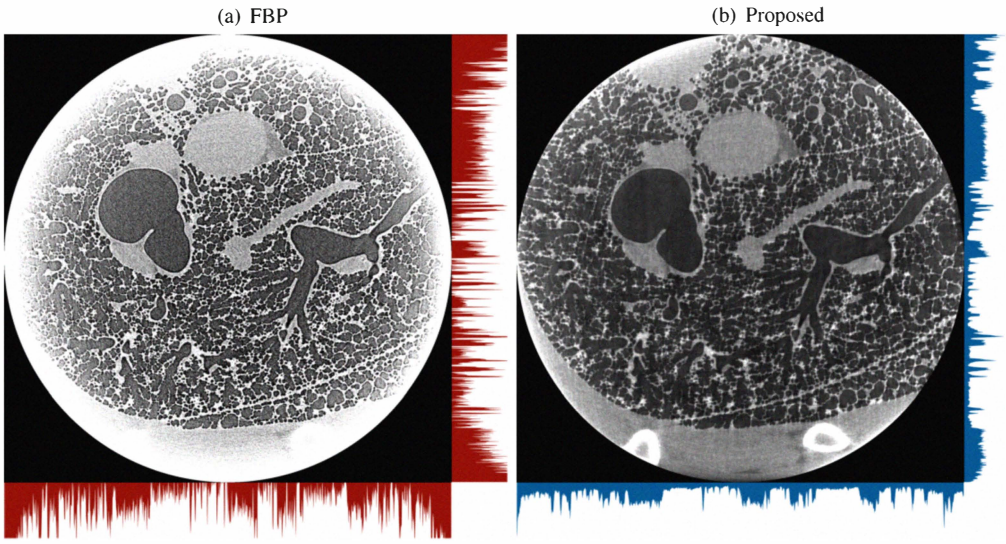

Fig. 4. Reconstructed images using FBP algorithm (left) with strong DC-shift and low-frequency artifacts and using the proposed iterative method (right) show a significant artifact reduction. Central horizontal and vertical profiles are also shown.

[4] H. Kudo, M.Courdurier, F.Noo and M.Defrise, "Tiny *a priori* knowledge solves the interior problem in computed tomography," *Phys. Med. Biol.*, vol. 53, pp. 2207-2231, 2008.

[5] X. Pan, Y. Zou and D. Xia, "Image reconstruction in peripheral and central regions-of-interest and data redundancy," *Med. Phys.*, vol. 32, pp. 673-684, 2005.

[6] E. Rashed and H. Kudo, "Region-of-Interest reconstruction from truncated projection data under blind object support", *Conference Record of 2008 IEEE Medical Imaging Conference*, Paper No. M03-7, 2008.

[7] Y. Ye, H. Yu, Y. Wei and G. Wang, "A general local reconstruction approach based on a truncated Hilbert transform," *Int. J. Biomed. Imaging*, vol. 2007, Article ID 63634 (8 pp), 2007.

[8] J. A. Fessler and A. O. Hero, "Penalized maximum-likelihood image reconstruction using space-alternating generalized EM algorithms," *IEEE Trans. Image Process.*, vol.4, pp. 1417-1429, 1995.

[9] http://www.spring8.or.jp